

National Institute of Technology Calicut
Department of Computer Science and Engineering

Winter Semester 2025-26
Course Plan for CS4033E: Computer Security

Credits: 3

Class: B. Tech. (CSE) Sem.: VI

Slot: D2

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Course Outcomes

- CO1: Demonstrate the design of modern cryptosystems
- CO2: Design cryptographic protocols
- CO3: Analyze and evaluate security of computer networks

Syllabus

Review of Cryptography Fundamentals - Symmetric and Asymmetric Cryptosystems, Cryptographic Hash, Digital Signatures.

Cryptographic Protocols for Authentication - Authentication Protocols: One way and Mutual Authentication, Challenge Response protocols, Needham Schroeder protocol. Interactive proof systems, Zero Knowledge Proof systems.

Network Security - Security at different layers – IPSec / SSL-TLS. Cloud security.

Principles of Secure Design - Software vulnerabilities - Buffer and stack overflow, Phishing. Malware - Viruses, Worms and Trojans. Security problems in network domain - Defense Mechanisms.

Reference Books

- B. Menezes, Network security and Cryptography, Cengage Learning India, 2010.
- B. A. Forouzan and D. Mukhopadhyay, Cryptography and Network Security, 2/e, Tata McGraw Hill, 2010.
- W. Stallings, Cryptography and Network Security - Principles and Practice, 8/e, Pearson India, 2023.
- D. Gollmann, Computer Security, 3/e, John Wiley and Sons Ltd., 2006.
- C. Kaufman, R. Perlman, and M. Speciner, Network Security, Pearson India, 2017.

Evaluation Scheme

Mid-Term : 30
Continuous Evaluation : 30 (Option-1: Three Quizzes (3 * 10) or Option-2: One Quiz (10) and One Security Attack assignment - Demonstrable (20))
End semester : 40

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Grading Policy

- Grading will be *relative*
- Deadline for submission of assignment and quizzes should be adhered to
- Makeup examination for quizzes and exams will be given only in genuine cases where written permission from the FA/HoD is obtained. There WILL not be any make-up quiz
- All issues regarding valuation of tests, quizzes, etc., must be resolved within two days after the marks are announced

Standard of Conduct

Each student is expected to adhere to high standards of ethical conduct, especially those related to cheating. Any academic dishonesty will result in zero marks in the corresponding exam or quiz and will be reported to the department council for record keeping and for permission to assign F grade in the course. CSE Department policy on academic integrity is available at: <http://minerva.nitc.ac.in/cse/sites/default/files/attachments/news/Academic-Integrity.pdf>

Security Attack assignment – Evaluation Details (20 marks)

- Individualistic and Demonstrable
- Submit the title of the attack vector to be demonstrated
- An implementation of a concept with demonstrable outcome is expected
- Two stages evaluation
 - Evaluation-1: (10 marks) Date: TBA
 - Literature Survey
 - Revised/Final Specification and Design
 - Clarity on Input/ Output
 - Progress of Implementation, if any
 - Evaluation-2: (10 marks) Date: TBA
 - Technical explanation
 - Implementation and Demonstration
 - Report submission
